Smaller Jets and Structured Arithmetic for Uniform Matrix Hitting Points in Positive Characteristic

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Abstract

We examine the multiplicative-shift construction in *Uniform Matrix Hitting Points in Every Positive Characteristic*. Retaining all initial orders in its Casoratian argument replaces the source cutoff $1 + n\binom{m}{2}$ by $m + (n-1)\binom{m}{2}$. An exact analysis of increasing matrix-index chains then replaces its coefficient precision by $1 + \binom{N}{2} + \max\{0, (n-1)(N-1) - n\}$; this is sharp for the specified common denominator, not a minimal hitting dimension. We prove division-free coefficient recurrences, a streamed letter action, and an exact ordered formula evaluator, with field-operation costs distinguished from binary-prime input and mandatory dense-output costs. Classical series-parallel branching programs further reduce a conservative state budget. The nonvanishing argument before either truncation is reproduced with its separate pole-orbit and finite-system hypotheses. Original Ram finite inversion supplies a useful coefficient-algebra comparison, but neither factorial derivatives nor complex positivity is transported to a finite field. The results are refinements of this specified construction, not an independent general PIT breakthrough or a priority claim. Contribution assessment: C3/5, explicit smaller truncation budgets and structured arithmetic give notable source-specific improvements while retaining the source's independent word-separation input.

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1 The model and the proof dependencies

Let p be a promised prime, F any field of characteristic p, and $n, s \geq 1$. A formula is a binary tree with ordered multiplication gates, addition gates, and leaves in $\{x_1, \ldots, x_n\} \cup F$. Its size counts all vertices. No division or subexpression sharing is allowed. Its value is the formal element of $F\langle x_1, \ldots, x_n\rangle$, not a function on the finite set $\mathbb{F}_p^n$. The machine constructing a tuple receives p in binary and n, s in unary; it receives neither the formula nor its coefficients. The same output over $\mathbb{F}_p$ must work over every such F.

The source [1] asserts a deterministic polynomial-bit-time singleton matrix substitution of dimension $O(n^3 s^6)$. This report reads its entire cached scientific manuscript, but distinguishes checking its displayed arguments from independent certification of the surrounding literature or the general derandomization problem. The dependency chain is

$$\begin{aligned} \text{pole orbits} &\implies \text{word independence} \\ &\implies \text{finite shift system and order bound} \\ &\implies z\text{-jet} \implies t\text{-jet} \implies \text{faithful matrices.} \end{aligned}$$

No last arrow proves an earlier arrow. In particular a faithful representation of a finite coefficient ring cannot establish that a nonzero formal word combination survived truncation.

Source file	Mathematical role	Treatment here
`introduction.tex`	Uniform quantifiers and encoding	Retained, including arbitrary extension coefficients
`words.tex`	Infinite-order shift, poles, word independence	Reproduced as the upstream source argument
`systems.tex`	Path coefficients and upper-unitriangular H	Reproduced; classical state budget refined
`order.tex`	Casoratian and minors over $K[[z]]$	All initial orders retained for a smaller bound
`construction.tex`	Two truncations, denominator and regular representation	Exact degree and arithmetic refinements proved

The main file, bibliography and README were also read. The five scientific files have 906 lines; the eight cached texts have 1122 lines. External references are not thereby read in full. Companion characteristic-zero reports and the parent's finite checks are not evidence counted as this author's proofs or tests.

2 What the private finite algebra does and does not supply

The original TR1 polynomial passage [6], lines 366–530, derives finite Taylor coefficients and, for a normalized *fixed* kernel, the matrix

$$A_{(a,b),(u,v)} = \binom{u}{a}\binom{v}{b}(-1)^{u+v-a-b}\mu_{u-a,v-b}, \qquad u \geq a, v \geq b.$$

Ordered by total degree this has identity diagonal blocks. The freshly read Volume 1 proof [7], lines 1–112, makes the corresponding finite nilpotent inversion precise: if $S = a_0 I + Q$, $Q^{D+1} = 0$, and

$a_0 \neq 0$, then

$$S^{-1} = a_0^{-1} \sum_{j=0}^{D} (-a_0^{-1} Q)^j. \tag{1}$$

Its implementation supplement, read in full, distinguishes derivative arrays from Taylor coefficients and checks the truncated reciprocal coefficient by coefficient. This is finite algebra, not an infinite Neumann convergence assertion.

Two restrictions are essential. TR1's broader forward preservation claim at lines 393–420 treats smooth spatially varying moment coefficients as though they could not increase degree. That inference is invalid without degree constraints on the moments. For example, integrating $f(x-t)$ against the normalized point measure $\delta_{x-x^2}(t)$ yields $f(x^2)$. The fixed-kernel statement is retained; the unrestricted one is not used. Second, $D^j/j!$ is not an operation available for arbitrary j in characteristic p. Use Hasse coefficients, defined by

$$f(x+z) = \sum_{j \geq 0} H_j f(x) z^j, \qquad H_j(fg) = \sum_{a+b=j} H_a f \, H_b g.$$

These follow directly by multiplying polynomials over any field. For $f = x^p$, ordinary $f' = 0$ but $H_p f = 1$.

The later private comparative report TR-item-317 [8], freshly read in full, supplies the useful warnings that an invertible coefficient change R requires transported multiplication $u \star v = R(R^{-1}u \, R^{-1}v)$, and that a real or complex positive Gram test cannot be asserted over a finite field. Already in $\mathbb{F}_5$ the nonzero vector $(1, 2)$ has sum of squares zero. This later report is not an original proof of the source's shift-word independence. Here we keep ordinary coefficient multiplication in a genuine quotient algebra, and use (1) only in its characteristic-free nilpotent form. No real moment measure, analytic norm, or positivity is required.

3 The source's nonvanishing mechanism before truncation

Put $K = F(t)$ with t transcendental, $a_i = (1+t)^i$, and $\sigma g(z) = g(tz)$. Then $K((z))^\sigma = K$: a coefficient at a nonzero integer power z^r would satisfy $(t^r - 1)c_r = 0$. Define

$$b_i(z) = \frac{z}{1 - a_i z}, \qquad (\sigma - 1) J_i g = b_i g, \qquad (J_i g)(0) = 0. \tag{2}$$

For $g \in K[[z]]$ this uniquely determines $J_i g$ by division by $t^r - 1$ for $r \geq 1$, and raises its z-order by at least one. Set $h_\varnothing = 1$, $h_{iw} = J_i h_w$. Thus letters prefix words while the rightmost operator acts first.

Lemma 3.1 (Source pole obstruction and word independence)**.** *The elements h_w, over all finite words, are linearly independent over $K(z)$. Consequently $f(J_1, \ldots, J_n)1 \neq 0$ for every nonzero formal $f \in F\langle x_1, \ldots, x_n \rangle$.*

Proof. The poles $\alpha_i = (1+t)^{-i}$ have disjoint infinite multiplicative t-orbits. Indeed $(1+t)^{i-i'} = t^a$ implies $a = 0$ by the valuation at $t = 0$, and $i = i'$ by that at $t = -1$. These are integer orders of rational functions, not integers reduced modulo p.

If $\sum_i c_i b_i = \sigma R - R$, $c_i \in K$, $R \in K(z)$, consider one such orbit. A nonempty finite set of poles of R on it has extreme indices $j_{\min}, j_{\max}$. The difference has surviving poles at $j_{\min} - 1$ and $j_{\max}$, two different points. The left side has at most one pole on that orbit. If R has no poles there it cannot produce one. It follows that every $c_i = 0$. Poles of R elsewhere do not affect this argument.

Suppose a finite rational relation among the h_w exists. Choose one with least maximal word length ℓ, then fewest nonzero coefficients of length ℓ, and normalize one such coefficient to 1. There is no relation of length zero. Apply σ and subtract the original relation. The coefficient at h_w in the result is

$$\widetilde{R}_w = \sigma R_w - R_w + \sum_i b_i \sigma R_{iw},$$

where missing coefficients are zero. The chosen top coefficient disappears and no new top coefficient occurs. Minimality therefore forces every $\widetilde{R}_w$ to vanish. At length ℓ this gives $R_w \in K$, using the Laurent-series fixed field. At length $\ell - 1$ it gives $\sum_i b_i R_{iw} = \sigma(-R_w) - (-R_w)$. The pole obstruction forces all top coefficients to vanish, a contradiction. The last claim is the resulting independence applied to the finitely many words of f. □

This is the source's substantive upstream proof, not a new consequence of Ram inversion. It uses a transcendental shift before reduction modulo t^E. After that reduction the map $z \mapsto tz$ is not invertible, so the pole-orbit or fixed-field argument must not be repeated there.

An acyclic path program with m vertices and scalar or variable edge labels has strictly upper triangular adjacency matrices $E_0, X_1, \ldots, X_n$ in a topological ordering. Parallel variable edges are counted in F. Let $S = (I - E_0)^{-1}$, a terminating sum, and set

$$u = e_{\mathrm{src}}^T S, \qquad B_i = X_i S, \qquad v = e_{\mathrm{snk}}. \tag{3}$$

The coefficient of $w = i_1 \cdots i_r$ is $uB_{i_1} \cdots B_{i_r} v$: each S accounts for scalar paths between consecutive variable edges. All B_i are strictly upper triangular, so products of length m vanish. Consequently

$$H = \sum_w B_w h_w \in \mathrm{GL}_m(K[[z]]), \quad H(0) = I, \qquad \sigma H = AH, \quad A = I + \sum_i b_i B_i = P/Q, \tag{4}$$

where $Q = \prod_i (1 - a_i z)$, $Q(0) = 1$, and both P and Q have z-degree at most n. In particular $uHv = f(J)1$.

4 Retaining the whole initial-order flag

Theorem 4.1 (Sharpened source Casoratian bound)**.** *Let K be any field and $\tau \in K^\times$ have infinite multiplicative order. Let σ fix K and send z to τz. Suppose $H \in \mathrm{GL}_m(K[[z]])$, $\sigma H = (P/Q)H$, $Q(0) \neq 0$, and $\deg P, \deg Q \leq \nu$. For a constant row u let k be the dimension over K of the span of the components of uH. If $y = uHv \neq 0$, with v constant and $\nu \geq 1$, then*

$$\mathrm{ord}_z\, y \leq (\nu - 1)\binom{k}{2} + k - 1 \leq (\nu - 1)\binom{m}{2} + m - 1. \tag{5}$$

If $\nu = 0$, necessarily $k = 1$ and $\mathrm{ord}_z\, y = 0$.

Proof. The finite-dimensional span admits a basis $w_1, \ldots, w_k$ with distinct initial orders $0 \leq c_1 < \cdots < c_k$, by successive elimination of leading coefficients. Any nonzero constant combination has initial order one of the c_i, hence at most c_k. If the initial coefficient of w_i is λ_i, the Casoratian determinant with rows $j = 0, \ldots, k-1$ and columns $i = 1, \ldots, k$ has leading term

$$z^{\sum_i c_i} \Big(\prod_i \lambda_i \Big) \det[(\tau^{c_i})^j]_{j,i}.$$

The last determinant is a nonzero Vandermonde, including in characteristic two. Its order is therefore $\sum_i c_i$.

Define $R_0 = u$, $R_{j+1} = \sigma(R_j)P/Q$. Then $\sigma^j(uH) = R_j H$. Writing

$$D_j = \prod_{e=0}^{j-1} Q(\tau^e z), \qquad R_j = N_j/D_j$$

gives $N_{j+1} = \sigma(N_j)P$ and $\deg N_j \le j\nu$. Stack $R_0, \dots, R_{k-1}$ into a $k \times m$ matrix M. Each nonzero k-minor of MH is a Casoratian of k independent components of uH, so has order $\sum_i c_i$; dependent choices give zero, and an independent choice exists. The minimum order of the maximal minors is unchanged by right multiplication by H. Cauchy–Binet gives one inequality because all minors of H lie in $K[[z]]$; applying it to H^{-1} gives the other.

A nonzero k-minor of M has unit denominator $\prod_{j<k} D_j$ and numerator of degree at most $\nu \sum_{j<k} j = \nu\binom{k}{2}$. Thus $\sum_i c_i \le \nu\binom{k}{2}$. Crucially,

$$\sum_{i<k} c_i \ge 0 + 1 + \cdots + (k-2) = \binom{k-1}{2}.$$

Subtracting proves the first bound in (5). Its right side increases from k to $k+1$ by $(\nu-1)k+1 > 0$. When $\nu = 0$, the sum of the distinct nonnegative orders is zero, forcing $k = 1$ and $c_1 = 0$. □

The improvement is retention of information already present in the source proof, not a newly invented determinant method. It does not claim optimality among all shift systems or all hitting constructions. For (4) it permits

$$N = N(n,m) := m + (n-1)\binom{m}{2}, \tag{6}$$

in place of $N_0 = 1 + n\binom{m}{2}$. Their difference is $\binom{m-1}{2}$. For $m = 1$, $N = 1$ and only a constant path output is possible. For $n = 1$, $N = m$, rather than a quadratic budget in m; univariate identity testing already has simpler classical solutions, so this is an improvement of this construction, not a new univariate result.

4.1 An optional classical reduction of the formula state budget

Proposition 4.2 (Series-parallel bookkeeping)**.** *A formula with q multiplication gates has a scalar/variable-edge acyclic path program on exactly $q+2$ vertices. Therefore a uniform budget for all size-at-most-s formulas is*

$$m_s = \left\lfloor \frac{s-1}{2} \right\rfloor + 2. \tag{7}$$

Proof. A leaf is an edge with distinct source and sink. Preserve the invariant that the source has no incoming edge and the sink no outgoing edge. For a sum, take disjoint child programs and identify their sources and their sinks. All paths remain in a single child, and the vertex count is $m_1 + m_2 - 2$. For an ordered product, identify the first child's sink with the second child's source; every path traverses the first and then the second, and the count is $m_1 + m_2 - 1$. These operations preserve acyclicity and the terminal invariant; parallel edges are allowed. Induction gives $q+2$. A binary tree of at most s vertices has at most $\lfloor (s-1)/2 \rfloor$ internal vertices, bounding q. □

This is classical series-parallel composition, explicitly present in the nonhomogeneous reduction underlying Raz–Shpilka [2]; the exact count is useful here but receives no independent novelty credit. The source's $m \le 2s$ remains valid and may be retained for a comparison isolating Theorem 4.1. The generator can use (7) without inspecting the hidden formula.

5 Exact denominator degree and finite-ring descent

In the column-coefficient basis $1, z, \dots, z^{N-1}$, the truncation of J_i is the strictly lower triangular matrix

$$(U_i)_{r,q} = \frac{(1+t)^{i(r-q-1)}}{t^r - 1} \quad (0 \le q < r < N), \qquad (U_i)_{r,q} = 0 \quad (q \ge r). \tag{8}$$

It follows directly by taking $[z^r]$ in (2). Raising z-order makes (z^N) invariant under every J_i, so ordered products descend correctly, and $f(U)e_0$ is $f(J)1$ modulo z^N. Lemma 3.1 and Theorem 4.1 imply that this column is nonzero for the bounded path-program family when N is (6).

Set

$$D_N(t) = \prod_{r=1}^{N-1} (t^r - 1), \qquad E = 1 + \binom{N}{2} + \max\{0, (n-1)(N-1) - n\}. \tag{9}$$

Empty products are 1, so $N = 1$ gives $D_N = 1$ and $E = 1$.

Theorem 5.1 (Sharp degree for this common denominator)**.** *For every $f \in F\langle x_1, \dots, x_n\rangle$, with no formula-size restriction, $D_N f(U)$ is a polynomial matrix in t of degree at most $E - 1$. This degree bound is attained by some f and some entry. This is sharpness for the fixed multiplier D_N, not a claim that D_N is a least common denominator or that E is a least hitting precision.*

Proof. For a nonempty word of length k a nonzero entry has a strictly increasing chain $q = j_0 < j_1 < \cdots < j_k = r < N$. Its contribution before clearing is

$$\frac{(1+t)^A}{\prod_{a=1}^{k} (t^{j_a} - 1)}, \qquad A = \sum_{a=1}^{k} i_{k-a+1}(j_a - j_{a-1} - 1) \le n(r - q - k).$$

The reversed letter index is required by ordered matrix multiplication. The denominator divides D_N as a product of distinct indexed factors; pairwise coprimality of those factors is neither true nor required. The degree after clearing is

$$\begin{aligned} \binom{N}{2} - \sum_{a=1}^{k} j_a + A &\le \binom{N}{2} - r + n(r - q - k) \\ &\le \binom{N}{2} + (n-1)r - n \le \binom{N}{2} + (n-1)(N-1) - n. \end{aligned}$$

The empty word contributes $D_N I$, of degree $\binom{N}{2}$. Words of length at least N vanish, and summing entries cannot raise degree. This proves the bound, including arbitrary extension-field coefficients.

If the maximum in (9) is zero, $f = 1$ attains it. Otherwise $N \ge 2$ and the $(N-1, 0)$ entry of $f = x_n$ has cleared numerator $D_N(1+t)^{n(N-2)}/(t^{N-1} - 1)$, of exactly the displayed degree. Its leading coefficient is 1 in every characteristic, so no reduction modulo p lowers that degree. □

The source precision is $E_0(N) = 1 + \binom{N}{2} + n(N-1)$. The new proof retains the degree removed by the actual denominators instead of discarding it. It remains valid when p divides an index r and $t^r - 1$ has multiple roots. At $t = 0$ every factor is still a unit.

Theorem 5.2 (Ordered descent with the refined budgets)**.** *Let N be (6), E be (9), and $R_F = F[t]/(t^E)$. Reduce (8) into $\mathrm{Mat}_N(R_F)$, obtaining V_i. Let ρ_F be multiplication on the coefficient basis $1, t, \ldots, t^{E-1}$ and apply it entrywise to obtain T_i of size $d = NE$. The T_i have entries in $\mathbb{F}_p$, independent of F, and for every nonzero path-program output with at most m vertices,*

$$f(T_1, \ldots, T_n)(e_0 \otimes 1) \neq 0. \tag{10}$$

Using $m = m_s$ gives a uniform tuple for all size-at-most-s formulas.

Proof. The nonzero column $f(U)e_0$ has a nonzero polynomial numerator column $D_N f(U)e_0$ of degree less than E. Reduction modulo t^E cannot annihilate that column. Since $D_N(0) \neq 0$, its image is a unit, so $f(V)e_0 \neq 0$. Reduction from the local ring of rational functions regular at zero is a unital algebra homomorphism, preserving all ordered products and arbitrary constants in F.

Multiplication by $c = \sum_{j<E} c_j t^j$ has matrix

$$\rho_F(c)_{a,b} = \begin{cases} c_{a-b}, & a \geq b, \\ 0, & a < b, \end{cases} \qquad 0 \leq a, b < E. \tag{11}$$

Composition is multiplication in R_F, not entrywise multiplication. It is faithful because multiplication by c sends 1 to c. Blockwise application is therefore a faithful unital representation of $\mathrm{Mat}_N(R_F)$. Moreover it commutes with extension from $\mathbb{F}_p$ to F and sends a constant α to αI. Applying the represented matrix to $e_0 \otimes 1$ recovers exactly the nonzero coefficient arrays of $f(V)e_0$. This proves (10). □

The hitting consequence is conditional on the upstream source mechanism as explicitly reconstructed in Section 3; the local degree and homomorphism proofs do not replace it. No ordinary derivative or factorial inverse is used anywhere in this descent.

5.1 A characteristic-two example and a compression boundary

For $N = 4$, a direct ordered multiplication gives

$$(U_1U_2 - U_2U_1)_{3,0} = \frac{((1+t) - (1+t)^2)t}{(t^3-1)(t^2-1)} = \frac{-t^2}{(t^3-1)(t-1)}. \tag{12}$$

All its other entries are zero. In characteristic two this remains nonzero and has t-order 2. Thus truncation modulo t^2 loses this commutator, whereas modulo t^3 detects it. More generally all the U_i coincide after setting $t = 0$, so that scalar specialization cannot replace the proved precision bound. This example is not itself a uniform cutoff theorem.

A row-dependent quotient precision must also respect the maps. Since each nonzero entry in (8) is a unit, a map from a component $F[t]/t^{E_q}$ to $F[t]/t^{E_r}$ induced by that entry is well-defined only if $E_q \geq E_r$. Increasing precision down the rows without this condition fails: the representative t^{E_q} of zero maps to a nonzero element if $E_q < E_r$. Entrywise degree estimates alone do not justify an adaptive direct-sum quotient.

Proposition 5.3 (Classical full-algebra size obstruction)**.** *A faithful unital F-algebra representation of $\mathrm{Mat}_N(F[t]/t^E)$ on a finite-dimensional F-space has dimension at least NE.*

Proof. The diagonal matrix units split the space into N summands. The off-diagonal units give inverse isomorphisms between every pair, so each has the same dimension r. The central element tI acts compatibly on the summands as a nilpotent operator A with $A^E = 0$. Faithfulness forces

$A^{E-1} \neq 0$, since $t^{E-1}e_{11} \neq 0$ in the algebra. A vector with $A^{E-1}v \neq 0$ gives E independent vectors $v, Av, \ldots, A^{E-1}v$ (apply the highest suitable power of A to a putative relation). Hence $r \geq E$ and the whole dimension is $Nr \geq NE$. This also includes $E = 1$. □

This elementary matrix-unit argument explains the cost of preserving the *entire* coefficient algebra. It is not a lower bound for hitting only the bounded formula family; a smaller different hitting construction is not excluded.

6 Division-free arithmetic and verifiable structured action

Let $\beta_e(A) = [t^e](1+t)^A$, interpreted in $\mathbb{F}_p$, with $\beta_e = 0$ for $e < 0$ or $e > A$. Pascal's recurrence constructs these coefficients using additions, even when $A \geq p$. No computation of $A!/(e!(A-e)!)$ in the field is legitimate in that range.

Proposition 6.1 (Coefficient recurrence and certificate)**.** *For $r \geq 1$ the coefficients c_e of $(1+t)^A/(t^r-1)$ modulo t^E satisfy uniquely*

$$c_e = -\beta_e(A) + c_{e-r} \quad (0 \leq e < E), \qquad c_j = 0 \ (j < 0). \tag{13}$$

Equivalently $c_e = -\sum_{j=0}^{\lfloor e/r \rfloor} \beta_{e-rj}(A)$. Checking (13) at all e certifies the exact truncated inverse product, not just a numerical residual.

Proof. Compare coefficients in $(t^r-1)\sum_{e<E} c_e t^e = (1+t)^A \bmod t^E$. The coefficient of c_e is -1, a unit even when $p = 2$. Forward substitution proves existence, uniqueness and the sum formula. □

This is the coefficient-residual principle in the private finite inversion proof specialized to the source denominator; the reciprocal identity itself is classical. The useful computational gain is that its denominator has only two terms.

Theorem 6.2 (Pre-output generation cost)**.** *For $m \geq 2$, all coefficient arrays of $V_1, \ldots, V_n$ can be generated in*

$$O(n^2NE + nN^2E) = O(nN^2E) \tag{14}$$

additions and subtractions in $\mathbb{F}_p$ on an indexed-array arithmetic model. The arrays use $O(nN^2E)$ field elements. Materializing the specified dense matrices requires nN^2E^2 entry slots, and must be accounted for separately.

Proof. For each i generate successive Pascal rows from exponent 0 through $i(N-2)$, truncated at E, by repeated multiplication by $1+t$. Each step costs $O(E)$ additions. Retain rows at exponents ig for $0 \leq g \leq N-2$. The total is $O(n^2NE)$. The row for $A = i(r-q-1)$ is shared by every entry with the same gap. Apply (13) in $O(E)$ operations to each of the fewer than $nN^2/2$ nonzero entries. For (6), $m \geq 2$ implies $N \geq n+1$, so the first term is absorbed by the second. Equation (11) writes each block by copying coefficients and zeros; it does not require a convolution or another inversion. For $m = 1$ the output consists only of 1×1 zero matrices and is immediate. □

Compared with the source's deliberately coarse schoolbook polynomial products, this avoids an extra E-scale product for each entry. It does not assert that this is the best possible polynomial arithmetic, or that dense output becomes cheap merely because the entries are structured.

6.1 Bit encoding and an ordinary machine

Write $L = \lceil \log_2 p \rceil$ and $B = n + s + L + 1$. With either the source state budget or (7), $N = O(B^3)$, $E = O(B^6)$ and $d = NE = O(B^9)$. A residue addition or subtraction with conditional reduction costs $O(L+1)$ bit operations on two explicitly stored residues. There is no loop of length p. Formula coefficients in an arbitrary extension field are not read by the generator.

For clarity, (14) is an indexed-array operation count, not an unproved random-access Turing bound. A completely elementary sequential simulation stores the at most $W = O(nN^2E)$ records on work tapes and retrieves or updates any indexed record by a full scan. The record and counter lengths are $O(L + \log B)$. If $T = O(nN^2E)$ is the arithmetic count, this gives the conservative bound

$$O((TW + nd^2W)(L + \log B)) \tag{15}$$

including even a full scan for each output entry. This is polynomial in the stated input length. Sequential record ordering can improve it; no optimal Turing exponent is claimed here. On an indexed-array model, expansion uses $O(nd^2)$ copying operations after the arithmetic stage. An explicit dense binary listing has length at most $O(\log d + nd^2(L+1))$, and at least $\Omega(nd^2)$ entry delimiters or slots in the prescribed format. Fixed-width L-bit residues give $\Theta(nd^2L)$ data bits; with variable-width encodings zeros need not cost L bits. These conventions prevent an erroneous output lower bound or a conflation of field-operation and bit complexity.

6.2 A streamed action and its exact residual

Theorem 6.3 (Letter action without dense blocks)**.** *For $v = (v_0, \dots, v_{N-1})^T \in R_F^N$ put $s_0 = 0$, $y_0 = 0$, and*

$$s_r = v_{r-1} + (1+t)^i s_{r-1}, \qquad (t^r - 1)y_r = s_r \pmod{t^E} \quad (1 \le r < N). \tag{16}$$

Then $y = V_i v$. The action costs $O(N(i+1)E)$ operations in F, with $O(NE)$ field-element storage including input and output.

Proof. Induction gives $s_r = \sum_{q<r} (1+t)^{i(r-q-1)} v_q$. Dividing by the unit $t^r - 1$ proves the formula. Multiplication by $(1+t)^i$ is i consecutive shift-add passes on an E-coefficient array, costing $O(iE)$. For the second equation, if $s_r = \sum s_{r,e} t^e$, use $y_{r,e} = -s_{r,e} + y_{r,e-r}$ with negative indices zero. This costs $O(E)$ and uses no field divisions. Iterate over the $N-1$ rows. □

Supplied s_r, y_r arrays can be checked by the two identities in (16). They uniquely determine $V_i v$. This certifies one action, not the assertion that a hidden polynomial is nonzero. Keeping that distinction is the finite-field counterpart of separating an exact Ram inverse residual from its application-level assumptions.

Corollary 6.4 (White-box ordered evaluation)**.** *Given a size-s formula over an effective field F of characteristic p, the vector $f(V)e_0$ can be computed in $O(sN(n+1)E)$ operations in F and $O(sNE)$ live field-element storage. With the proved uniform budgets it vanishes exactly when the formal polynomial is zero. Costs of arithmetic on the supplied extension-field coefficients must be specified separately.*

Proof. Define an action evaluator $\mathcal{A}(g, v)$. A scalar leaf multiplies v by that scalar, and a variable leaf uses Theorem 6.3. At a sum, add $\mathcal{A}(g_1, v)$ and $\mathcal{A}(g_2, v)$. At an ordered product use $\mathcal{A}(g_1, \mathcal{A}(g_2, v))$. Induction proves $\mathcal{A}(g, v) = g(V)v$. Each tree vertex is visited once, additions and scalar leaves cost $O(NE)$, and a variable leaf costs at most $O(N(n+1)E)$. Store at most one constant number of vectors per recursion level to obtain the stated live bound. Finally use (10). □

This is not a black-box evaluator: it reads the formula tree. Shared DAGs may require repeated actions on different input vectors, so the tree count cannot be asserted for unrestricted circuits. Deterministic white-box noncommutative PIT is already classical [2]; the result here is a structured way to apply this particular tuple. It neither changes the mandatory dense output of the uniform generator nor charges arbitrary F arithmetic as $O(L)$ bit operations.

7 Quantitative comparison, limits and further work

The following exact budgets keep the same path-program state bound m on both sides, isolating the order and degree refinements. They do not include the optional smaller formula state count.

		Source budgets			Refined budgets		
n	m	N_0	$E_0(N_0)$	d_0	N	E	d
1	6	16	136	2176	6	16	96
2	4	13	103	1339	10	53	530
2	6	31	526	16306	21	229	4809
3	4	19	226	4294	16	148	2368

For fixed $n \geq 2$ and $m \to \infty$, the ratio of these dimensions tends to $((n-1)/n)^3$. Both still have order m^6; no improved general exponent in the source's $O(n^3 s^6)$ theorem is claimed. For $n = 1$ the refined construction has order m^3 rather than m^6, but ordinary univariate coefficient methods are preferable. The degree refinement also has value at small m: at $N = 2$ it gives $E = 2$ independently of n, since the only nonzero entry is $1/(t-1)$.

What is retained and what is changed. No mathematical error in the read positive-characteristic source is required for these improvements. Its conservative budgets remain valid. The infinite-order shift, the pole obstruction, the order estimate before truncation, distinct-index denominator clearing, arbitrary extension-field coefficients, and ordered regular representation are retained. The local changes are the subtraction of the other initial orders, exact cleared degree, and structured arithmetic. The classical terminal identifications and full-algebra dimension obstruction clarify cost; they are not new PIT theorems. The only rejected proof inference discussed here is the unrestricted polynomial preservation for varying kernels in the earlier private TR1, not a claim that its correct fixed-kernel derivation fails.

Advantages. The refinements reduce concrete certificate dimensions without choosing a large finite-field specialization or factoring a polynomial. They work uniformly at $p = 2$ and need no factorial denominators. Exact recurrence residuals provide independently checkable local data. The streamed action uses the specific numerator/denominator pattern, rather than merely renaming a lower triangular matrix. Faithful coefficient multiplication keeps word order and extension coefficients intact.

Costs and limitations. The matrices can still be enormous, and dense output is not replaced by a succinct description under the source contract. The faithful representation lower bound applies only to the whole ring algebra. The smaller coefficient precision is sharp only for the chosen common denominator; it need not be sharp for the bounded formula class. Indexed arithmetic speedups are not automatically matching multitape bit-speedups. No empirical runtime improvement has been measured. Formula evaluation over an arbitrary field is an algebraic operation count until that field's

effective representation is supplied. No finite experiment establishes the all-words independence assertion or verifies every possible input.

Feasible extensions. One can certify a smaller component span k for a supplied white-box system and use the first bound in (5); a uniform hidden formula generator cannot assume that k is known. A different common denominator or a bound for only the first witness column may reduce E, but must be proved simultaneously for all allowed words and coefficients. An optimized sequential coefficient layout could improve the deliberately coarse bit bound (15). Any row-adaptive precision must define a module on which every ordered letter acts, as the counterexample in Section 5 shows. These are concrete further tasks, not promises of another general derandomization result.

8 Primary literature and the status of the comparison

Public searches were restricted to public titles and mathematical terms, with the research cutoff October 8, 2026. No private title or passage was sent to a public service. The cached target is the October 4 preprint at the pinned repository commit; the parent's captured-October-8 identity record authenticates bytes, not a live branch or a proof.

Raz and Shpilka [2] give deterministic polynomial-time *white-box* identity testing for noncommutative formulas and branching programs. Their author-hosted introduction and nonhomogeneous series-parallel construction were inspected, not the full paper's proof. Forbes and Shpilka [3] give quasipolynomial black-box hitting sets for known-order read-once oblivious branching programs and reductions from noncommutative models; the current arXiv abstract and metadata were checked. These establish the classical context, not the sharper budgets proved here.

The July 20, 2026 revision of Lakhani–Mukhopadhyay [4] describes a single integer matrix hitting tuple for sparse noncommutative polynomials over $\mathbb{C}$, with complexity controlled by sparsity. Its primary ECCC abstract and revision metadata were read; the PDF endpoint was unavailable to the browser tool and is not claimed as a full proof read. Sparsity, formula size, characteristic zero, and all positive characteristics are different parameters and domains. Arvind–Chatterjee–Mukhopadhyay [5] report quasipolynomial black-box testing for noncommutative rational formulas; the version-4 abstract was checked. Rational formulas with inverses are outside the present division-free model. None of these bounded literature reads establishes historical priority for the local inequalities, or independently certifies the source's stronger uniform singleton assertion. The present report supplies a source-specific mathematical comparison, not a claim that a general commutative circuit problem has been solved.

9 Verification and contribution assessment

The executable `_audit/M231-checks.py` uses exact integers, prime fields of characteristics 2, 3, 5 and 7, and $\mathbb{F}_4 = \mathbb{F}_2[u]/(u^2 + u + 1)$. The independent deterministic run passed 6870 cases, with a case meaning an entire indicated identity or instance, not each scalar assertion inside it:

Case group	Count
Streamed versus dense letter action, prime-field vectors	864
Complete inverse coefficient array and residual	2808
Toeplitz multiplication versus coefficient convolution	36
Streamed versus dense action over $\mathbb{F}_4$	72
Polynomial Casoratian versus Vandermonde product	224
Increasing-chain common-denominator degree	2330
Actual polynomial degree attainment modulo primes	140
Series-parallel path polynomial versus formula expansion	180
Ordered recursive action versus independent word expansion	180
Commutator and loss at insufficient precision	36
Total	6870

The inverse reference uses integer binomial coefficients and a direct geometric sum, independently of the Pascal/recurrence implementation. The action reference uses dense convolution. Formula checks expand words independently and compare path sums. Polynomial Casoratians are computed by permutation determinants and compared with products of differences in $\mathbb{F}_p[t]$. Chain tests check worst-letter degrees and separate exact polynomial attainments. These are manufactured finite checks, not a census of formulas, an experimental complexity measurement, or a substitute for the proofs. The parent's separately executed suites are not included in these counts.

The source-read ledger records literal displays, pre/post SHA-256 hashes, and unread complements. Original private reads total 406 lines: 165 of TR1, 112 of the revised Volume 1 chapter, and all 129 lines of its implementation supplement. The 434-line TR-item-317 was separately read in full as a later comparison. No other private book, original report, or external bibliography is claimed read in full. The public manuscript's complete cached text read does not amount to a full-corpus proof audit.

The assessment is **C3**, independently of catalogue applicability **L3**. The notable local increment is the combined reduction of the two truncation budgets with explicit sharpness scope and a genuinely structured coefficient/action algorithm. Finite-ring regular representations, Hasse coefficients, Vandermonde determinants, series-parallel branching programs and nilpotent reciprocals are classical tools, not credited as newly discovered results. A C4 or C5 claim would require substantially more than these refinements. In particular the report does not supply a different all-word separation method, a better general asymptotic matrix-dimension exponent, an unrestricted circuit algorithm, or a new uniform arithmetic claim for unspecified coefficient fields.

References

[1] OpenAI, *Uniform Matrix Hitting Points in Every Positive Characteristic*, October 4, 2026, cached source at commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Public source directory.

[2] R. Raz and A. Shpilka, *Deterministic polynomial identity testing in non-commutative models*, Computational Complexity 14 (2005), 1–19. Author-hosted paper.

[3] M. A. Forbes and A. Shpilka, *Quasipolynomial-time identity testing of non-commutative and read-once oblivious algebraic branching programs*, arXiv:1209.2408v2, 2013. Primary abstract and version history.

[4] F. Lakhani and P. Mukhopadhyay, *Hitting point for sparse noncommutative polynomials*, ECCC TR26-120, revision 2, July 20, 2026. Primary report record.

[5] V. Arvind, A. Chatterjee and P. Mukhopadhyay, *Black-box identity testing of noncommutative rational formulas in deterministic quasipolynomial time*, arXiv:2309.15647v4, July 11, 2025. Primary abstract and version history.

[6] M. Subbarao, original local TR1 polynomial derivations, `TR1-poly/tr1-2026-03-26-0001.tex`, literal lines 366–530. Private source, not publicly uploaded. Fixed-kernel results are distinguished from the variable-kernel limitation in Section 2.

[7] M. Subbarao, *Ram Calculus, Volume 1*, revised Chapter 2, `ch02-s-transform-rev2.tex`, lines 1–112, and `ch02-A-implementation.tex`, lines 1–129. Private original finite-moment and reciprocal derivations.

[8] M. Subbarao, private local `res-math/TR-item-317-astra.tex`, 434 lines, later comparative report. Hasse jets, transported products and measurement limitations; not a foundational source of the target's word independence.